

ECONOMY TOPIC 27 | DIGITAL AGRICULTURE, AGRITECH AND e-TECHNOLOGY

DATA CUTOFF: 10 SEPTEMBER 2026 | EVERY CURRENT CLAIM NEEDS DATE + STATUS
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[01 VALUE CHAIN]
DATA CAPTURE -> ID/REGISTRY -> ADVICE -> INPUT/CREDIT/INSURANCE
-> PRODUCTION -> STORAGE/LOGISTICS -> MARKET -> PAYMENT -> FEEDBACK
Outcome test: net income, risk, resource use, resilience and farmer control.

[02 TECHNOLOGY STACK]
Remote sensing observes | GIS analyses | GNSS/NavIC positions
IoT senses/exchanges | AI predicts | automated control acts
Drone capability = airframe + payload + software + trained operation.

[03 PRECISION AGRICULTURE]
Measured spatial/temporal variation -> variable treatment -> feedback
Generic digitisation merely converts/transmits information.
Efficiency per hectare may rebound into higher total resource extraction.

[04 PUBLIC ARCHITECTURE]
IDEA 2021 blueprint -> Digital Agriculture Mission approved 3 Sep 2024
Rs 2,817 crore outlay; Rs 1,940 crore central share
AgriStack + Krishi-DSS + crop-estimation/support systems.

ECONOMY TOPIC 27 - ASCII MASTER CONTINUED

[05 AGRISTACK]
Farmer Registry/Farmer ID + georeferenced village maps + Crop Sown Registry
Federated state/UT operation + common standards + authorised service interfaces
Farmer ID != land title != cultivation != guaranteed entitlement.

[06 CURRENT SNAPSHOT]
DA&FW page checked 10 Sep 2026: >10 crore IDs; ~5 lakh maps;
>30 crore plots surveyed in 650 districts. Page displayed no publication date.
Coverage != accuracy != adoption != welfare outcome.

[07 DATA GOVERNANCE]
Purpose + lawful basis + minimisation + security + retention + provenance
+ correction + human review + appeal + offline/assisted route
DPDP Rules notified 13 Nov 2025 with phased commencement.

[08 FARM FINANCE / RISK]
PM-KISAN: digital verification -> DBT -> physical correction where needed
KCC: verified data -> bank underwriting; not automatic credit
PMFBY: CCE/YES-TECH/WINDS -> claim; basis risk remains.

ECONOMY TOPIC 27 - ASCII MASTER CONTINUED

[09 MARKETS / BUSINESS]

e-NAM online layer + assaying + aggregation + logistics + settlement
FP0/shared service spreads fixed cost; unit economics decides survival
Network effects can improve matching and create lock-in.

[10 INCLUSION]

Connectivity -> device -> skill -> language -> trust -> effective use
Protect tenants, sharecroppers, women, smallholders and minor crops
Human extension complements digital delivery.

[11 ACCOUNTABILITY]

AI/score/model -> confidence + subgroup test + explanation + monitoring
Adverse decision -> notice -> assisted correction -> human appeal
Cyber resilience -> prevent -> detect -> contain -> recover -> learn.

[12 UPSC SYNTHESIS]

Problem -> tool -> named evidence -> mechanism -> weakest complement
-> exclusion/risk -> reform -> net outcome + qualification
Topic 29 owns wider mission-mode technology portfolios.